

#This code will be used to demonstrate theorem 6.17. We will demonstrate that $Q2 = ((w+z)^3 - y^3) \cdot x \cdot (-w^2 \cdot (z^3 - y^3) + z^2 \cdot (w^3 - y^3)) / (y \cdot (w+z)^2 \cdot ((x^3 - w^3) \cdot (z^3 - y^3) + (x^3 - z^3) \cdot (w^3 - y^3)))$ is decreasing along the γ_1 curve; (Part 1 – locating potential critical points)

#Define Q2;

$$Q2 := \frac{((w+z)^3 - y^3) \cdot x \cdot (-w^2 \cdot (z^3 - y^3) + z^2 \cdot (w^3 - y^3))}{y \cdot (w+z)^2 \cdot ((x^3 - w^3) \cdot (z^3 - y^3) + (x^3 - z^3) \cdot (w^3 - y^3))} :$$

#Apply the rational parametrization constructed in this chapter 4.

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$y = a - \frac{1}{a} + b - \frac{1}{b},$$

$$x = a - \frac{1}{a} - b + \frac{1}{b} \Big\} :$$

$$q2 := \text{simplify}(\text{expand}(\text{subs}(\text{parametrization}, Q2))) ;$$

$$\begin{aligned} q2 := & \left((a^8 b^4 + 2(2b^5 - b^3)a^7 + (7b^6 - 6b^4 + b^2)a^6 + 2(2b^7 - 6b^5 + b)a^5 + (b^8 - 6b^6 \right. \\ & + 4b^4 + 10b^2 - 1)a^4 + 2(-b^7 + 6b^3 - 2b)a^3 + (b^6 + 10b^4 - 5b^2)a^2 + 2(b^5 \\ & - 2b^3)a - b^4)(a-b)^2 \left(a^2 b^2 + \frac{1}{3} \right) \Big) \Big/ \left(2(ab+1) \left((b^8 - b^6 + b^4)a^{10} \right. \right. \\ & - \frac{b^2(b^8 - 2b^6 + 8b^4 - 2b^2 + 1)a^8}{2} + \frac{b(b^4 - 4b^2 + 1)(b^2 + 1)^3 a^7}{2} \\ & + \frac{(b^4 + 1)(b^8 + 8b^4 + 1)a^6}{6} - \frac{b(b-1)^2(b+1)^2(b^2 + 1)^3 a^5}{2} \\ & \left. \left. - \frac{b^2(b^8 + 2b^6 - 2b^4 + 2b^2 + 1)a^4}{2} - \frac{b^3(b^2 + 1)^3 a^3}{3} + b^6 a^2 - \frac{b^6}{3} \right) \right) \end{aligned} \quad (1)$$

#Let us denote by $D1 = \frac{da}{db}$. By the chain rule, the derivative of Q2 is zero if and only if

$$D1 := - \frac{\text{diff}(q2, b)}{\text{diff}(q2, a)} :$$

#Define r1

$$\begin{aligned} r1 := & \left(a^{32} b^{16} + b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} \right. \\ & + (72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}) a^{26} + (b^{32} + 8 b^{30} + 20 b^{28} \\ & - 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} \\ & - 157780 b^{12} - 32376 b^{10} + 1950 b^8 + 1672 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-1016 b^{26} \\ & - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} \\ & + 8 b^8 + 8 b^6) a^{10} + (-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} \\ & + 144 b^{12} - 72 b^{10} - 9 b^8) a^8 + (-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} \\ & + 72 b^{12}) a^6 + (-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} \\ & - 4 b^{14}) a^2 - 2560 \left(b^2 - \frac{1}{5} \right) b^7 (b^2 - 2) (b^2 + 1)^4 a^{23} + 2 b^4 (b^{24} + 36 b^{22} - 2636 b^{20} \\ & - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^8 - 1372 b^6 + 180 b^4 \\ & + 36 b^2 + 1) a^{12} - 4 b^2 (b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} \\ & + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^8 + 378 b^6 + 37 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} \\ & + 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} \\ & + 36360 b^{10} + 5211 b^8 - 2118 b^6 - 539 b^4 + 76 b^2 + 1) a^{18} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} \\ & - 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^8 - 348 b^6 + 3252 b^4 \\ & + 420 b^2 - 127) a^{20} + 8 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^8 \right. \\ & + 2369 b^6 - 699 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 9 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^8 - 328 b^6 \right. \\ & - 528 b^4 + \frac{4168}{9} b^2 + 1 \left. \right) b^8 a^{24} - 1024 b^{15} (b^2 + 1)^4 a^7 - 2560 \left(b^6 - \frac{8}{5} b^4 + \frac{6}{5} b^2 \right. \\ & - 1 \left. \right) b^{11} (b^2 + 1)^4 a^{11} - 1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 + 1)^5 a^9 + 512 b^7 (b^{10} - 19 b^8 + 42 b^6 \\ & - 9 b^4 - 10 b^2 + 1) (b^2 + 1)^4 a^{15} - 1536 \left(b^8 - b^6 + \frac{5}{3} b^4 - \frac{16}{3} b^2 + \frac{5}{3} \right) b^9 (b^2 + 1)^4 a^{13} \end{aligned}$$

$$\begin{aligned}
& + 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 + 11 b^2 + 1) (b^2 + 1)^4 a^{17} - 3584 b^7 \left(b^8 - \frac{52}{7} b^6 \right. \\
& + \frac{95}{7} b^4 - \frac{43}{7} b^2 + \frac{3}{7} \left. \right) (b^2 + 1)^4 a^{19} + 5632 \left(b^6 - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11} \right) b^7 (b^2 \\
& + 1)^4 a^{21} \Big) :
\end{aligned}$$

#Using the implicit function theorem, compute $\frac{da}{db} = D2$ in terms of a and b

$$D2 := - \frac{\text{diff}(r1, b)}{\text{diff}(r1, a)} :$$

#We construct the curve $D2 - D1$ where its zeros describe points where the derivative of $\left((w+z)^3 - y^3 \right) \cdot x \cdot \left(-w^2 \cdot (z^3 - y^3) + z^2 \cdot (w^3 - y^3) \right) \Big / \left(y \cdot (w+z)^2 \cdot \left((x^3 - w^3) \cdot (z^3 - y^3) + (x^3 - z^3) \cdot (w^3 - y^3) \right) \right)$ is zero

$$f := D2 - D1 :$$

#Calculate the numerator of f

$$\tilde{f} := \text{numer}(f) :$$

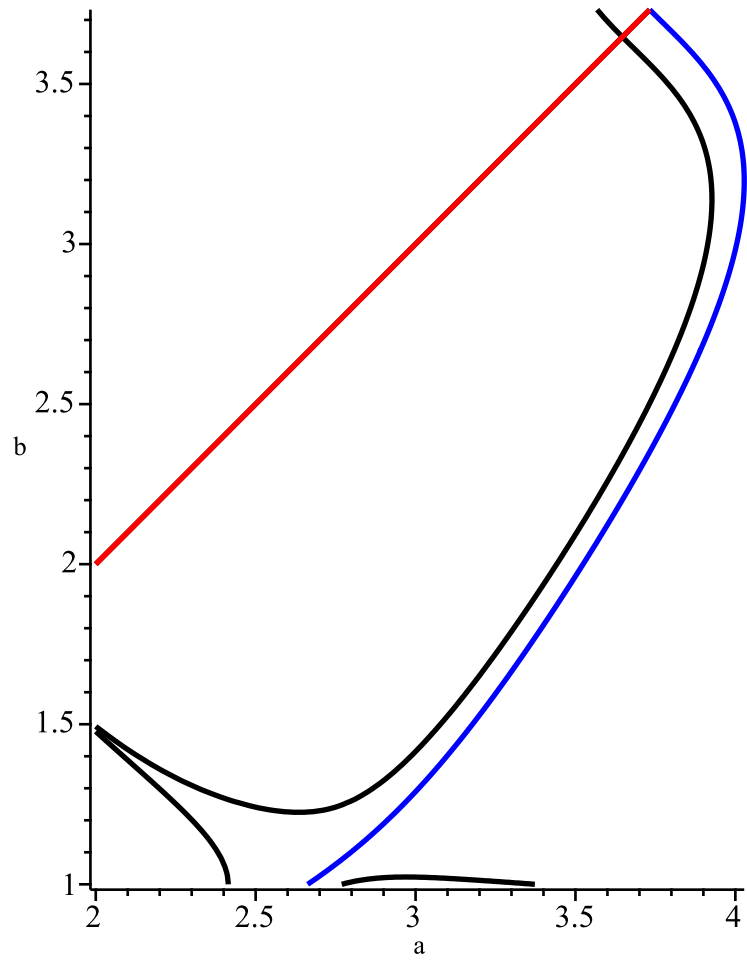
#Plot the curves $r1=0$ and $\tilde{f}=0$ in region $A1$.

$$r := a - b :$$

with(plots) :

```
implicitplot([ r1=0,  $\tilde{f}=0$ ,  $r=0$ ], a=2 .. 5, b=1 .. (2 + sqrt(3)),
  scaling=constrained,
  gridrefine=2,
  thickness=2,
  color=[blue, black, red],
  labels=["a", "b"],
  title="Graphic for  $r1=0$  and  $\tilde{f}=0$  in  $A1$ ");
```

Graphic for $r_1=0$ and $\tilde{f}=0$ in A1



#Let's build the DescartesSingChanges function that counts the number of sign changes between the coefficients of a given polynomial.

```
DescartesSignChanges := proc (polynomial, variable)
  local coeffs, nonzero_coeffs, s, i, n;

  # Get the list of coefficients from highest degree to constant term
  coeffs := [seq(coeff (polynomial, variable, i), i = degree (polynomial, variable) .. 0, -1)];

  # Remove zero coefficients (they do not count in Descartes' Rule)
  nonzero_coeffs := remove (c → c = 0, coeffs);

  # Count the number of sign changes
  n := nops (nonzero_coeffs);
  s := 0;
  for i from 1 to n - 1 do
    if signum (nonzero_coeffs[i]) * signum (nonzero_coeffs[i + 1]) = -1 then
      s := s + 1;
    end if;
  end do;

  return s;
end proc;
```

#Compute the resultant between $r1$ and $\tilde{f}$ with respect to a

$R1 := \text{resultant}(r1, \tilde{f}, a) :$

#Compute the factors of the resultant $R1$ and count how many there are

```
dgb := proc(f)
  if type(f, ``) then
    degree(op(1, f), b);
  else
    degree(f, b);
  end if;
end proc;
```

```
bfator := sort(
  [op(factor(R1))],
  (f, g) → dgb(f) < dgb(g)
);
```

$\text{nops}([op(factor(R1))]);$

15

(2)

#We list below the factors of the resultant $R1$, except for the factor 15 because it is too large. When necessary, we will use Sturm's algorithm to check if the factor cancel each other out within the interval $[1, 2 + \sqrt{3})$ where γ is defined.

$rb1 := bfator[1];$

$rb1 := 206294879089744981779124551495182122153863456743895446897623467852562432$ (3)

$rb2 := bfator[2];$

$rb2 := b^{1296}$ (4)

$rb3 := bfator[3];$

$rb3 := b^2 + 4b + 1$ (5)

$rb4 := bfator[4];$

$rb4 := b^2 - 4b + 1$ (6)

$rb5 := bfator[5];$

$rb5 := (3b^2 - 1)^5$ (7)

$rb6 := bfator[6];$

$rb6 := (b^2 + 1)^{494}$ (8)

$rb7 := bfator[7];$

$rb7 := b^4 - 4b^3 + 18b^2 - 4b + 1$ (9)

$$\begin{aligned}
rb8 &:= bfator[8]; \\
sturm(sturmseq(rb8, b), b, 1, 2 + \sqrt{3}); &\#number\ of\ roots\ in\ [1, 2 + \sqrt{3}) \\
rb8 &:= b^4 + 4b^3 + 18b^2 + 4b + 1 \\
0
\end{aligned} \tag{10}$$

$$\begin{aligned}
rb9 &:= bfator[9]; \\
sturm(sturmseq(rb9, b), b, 1, (2 + \sqrt{3})); &\#number\ of\ roots\ in\ (1, (2 + \sqrt{3})) \\
rb9 &:= (3b^4 - 6b^2 + 7)^5 \\
0
\end{aligned} \tag{11}$$

$$\begin{aligned}
rb10 &:= bfator[10]; \\
sturm(sturmseq(rb10, b), b, 1, 2 + \sqrt{3}); &\#number\ of\ roots\ in\ [1, 2 + \sqrt{3}) \\
rb10 &:= 65b^6 - 141b^4 + 195b^2 - 47 \\
0
\end{aligned} \tag{12}$$

$$\begin{aligned}
rb11 &:= bfator[11]; \\
sturm(sturmseq(rb11, b), b, 1, 2 + \sqrt{3}); &\#number\ of\ roots\ in\ [1, 2 + \sqrt{3}) \\
rb11 &:= (b^6 - 45b^4 + 3b^2 - 15)^2 \\
0
\end{aligned} \tag{13}$$

$$\begin{aligned}
rb12 &:= bfator[12]; \\
rb12 &:= (b^6 + 51b^4 + 3b^2 + 17)^2
\end{aligned} \tag{14}$$

$$\begin{aligned}
rb13 &:= bfator[13]; \\
sturm(sturmseq(op(1, rb13), b), b, 1, 2 + \sqrt{3}); &\#number\ of\ roots\ in\ [1, 2 + \sqrt{3}) \\
rb13 &:= (374017b^{48} - 1082256b^{46} - 2821284b^{44} + 38273024b^{42} - 137706774b^{40} \\
&\quad - 25011408b^{38} + 2222925500b^{36} - 11137697376b^{34} + 34666181511b^{32} \\
&\quad - 79729549344b^{30} + 140202896376b^{28} - 190367984256b^{26} + 204746625068b^{24} \\
&\quad - 177535429920b^{22} + 126486144024b^{20} - 74566977600b^{18} + 36572797215b^{16} \\
&\quad - 14813913168b^{14} + 4932365356b^{12} - 1327146624b^{10} + 292102794b^8 - 53430800b^6 \\
&\quad + 8765868b^4 - 983136b^2 + 45241)^{10} \\
0
\end{aligned} \tag{15}$$

$$\begin{aligned}
rb14 &:= bfator[14]; \\
sturm(sturmseq(op(1, rb14), b), b, 1, 2 + \sqrt{3}); &\#number\ of\ roots\ in\ [1, 2 + \sqrt{3}]; \\
rb14 &:= (4170153564935809983b^{156} - 347072952894808621950b^{154} \\
&\quad + 1002344030045036839125b^{152} + 34068413508632845696116b^{150} \\
&\quad + 1764554824835978005223463b^{148} - 17407565191158737689970946b^{146} \\
&\quad - 216048149459477528028020827b^{144} + 1956213362565512717830833816b^{142} \\
&\quad + 13829661974246732966859355485b^{140} - 162340076637417985627564578514b^{138} \\
&\quad + 49863654734008661388943836783b^{136} - 9900984422511556818979295917740b^{134} \\
&\quad + 111426996676113263095944985685901b^{132}
\end{aligned}$$

$$\begin{aligned}
& + 619860369954219114028234260172626 \, b^{130} \\
& - 8323582120659525168536263620251961 \, b^{128} \\
& - 14506644890019352043843463903049664 \, b^{126} \\
& + 486560265839604073196474197116043164 \, b^{124} \\
& - 3003570392540101865087764748741303160 \, b^{122} \\
& + 10762446129766802461316300362871686068 \, b^{120} \\
& - 20983044594507402382571162605982323056 \, b^{118} \\
& + 12357330403805598669075164837520603900 \, b^{116} \\
& + 22149057370604876964509659678447189624 \, b^{114} \\
& + 51063565384779734300946344512014119220 \, b^{112} \\
& - 343615675526010254474949876055978241376 \, b^{110} \\
& + 572304726582372003943456557444324658660 \, b^{108} \\
& - 301516206764056666239464248319853242472 \, b^{106} \\
& - 15866249334274782087646245588033249780 \, b^{104} \\
& - 1536436454887031164160118447625589712944 \, b^{102} \\
& + 4966509761132987187123479749849535262948 \, b^{100} \\
& - 7256092189817071284720504677788303708824 \, b^{98} \\
& + 5360558687346575145805290233225679935404 \, b^{96} \\
& - 4732008090683601101198916412388181802560 \, b^{94} \\
& + 14564401603684848684541945022041763295618 \, b^{92} \\
& - 35824619471723942177739204744538576767172 \, b^{90} \\
& + 54063291285060043601451079668741746632662 \, b^{88} \\
& - 58724410174258598649006211920296209780776 \, b^{86} \\
& + 61656051419883747450238738450383134563058 \, b^{84} \\
& - 90729161554414712574897887505486629206332 \, b^{82} \\
& + 153561585682384177912187927176499510824182 \, b^{80} \\
& - 220004269000606531937914985420979881281136 \, b^{78} \\
& + 243591856351288543568598444641772571843158 \, b^{76} \\
& - 205307506061458741821964537288964359204668 \, b^{74} \\
& + 128483772924084915191410765495527023342930 \, b^{72} \\
& - 55979376886560926403656740401524260235304 \, b^{70} \\
& + 14856122074723924124298860722562802725814 \, b^{68} \\
& - 3955205261570191183123912384193755367108 \, b^{66} \\
& + 6563805327143655203940509170540028090466 \, b^{64}
\end{aligned}$$

$$\begin{aligned}
& - 8833375032775702807948649709388563447360 b^{62} \\
& + 7013659464620873378334148195447331743852 b^{60} \\
& - 3540004390856713390340444922603703221912 b^{58} \\
& + 1058590461372895421771282653974865740708 b^{56} \\
& - 116614354222940452510620157173133006896 b^{54} \\
& + 12072987991564623810576421275785035212 b^{52} \\
& - 76388760308187455911174939910611108968 b^{50} \\
& + 81107980734204901315267408424455543716 b^{48} \\
& - 46121682190537227842779749333610917216 b^{46} \\
& + 17468220296985996262470337578279187572 b^{44} \\
& - 4829981885821823924585546414792734088 b^{42} \\
& + 1039795400253525481034862310322193468 b^{40} \\
& - 171791709178326296999845457915208048 b^{38} \\
& + 13032733636063165568183302585300468 b^{36} \\
& + 4173537101566204561731174651183240 b^{34} \\
& - 2023145351439782210888740989477924 b^{32} \\
& + 494978415415112481078763544152128 b^{30} - 92300599687783329815049589360713 b^{28} \\
& + 12005386098267790709457670903890 b^{26} - 182598334338368271586785687811 b^{24} \\
& - 310481324910885734882601124524 b^{22} + 79539059828434235052427832607 b^{20} \\
& - 13927305220149044325187283410 b^{18} + 1972979746709284943883794829 b^{16} \\
& - 160940106010701027705944424 b^{14} - 3620019864154595920678667 b^{12} \\
& + 279967648023527012610558 b^{10} + 130388126939757400959735 b^8 \\
& + 130963328751372183668 b^6 - 361585232998046982939 b^4 - 26091219170754511998 b^2 \\
& - 201545550647774641)^2
\end{aligned}$$

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(16)

#Finding possible candidates for zeros of the factor rb14

fsolve(op(1, rb14) = 0, b, 1 .. (2 + sqrt(3)));

1.378512042, 2.437789845, 3.465553247

(17)

#The factor rb15 has degree 612, so we will not show it, just compute how many zeros it has in the desired range.

rb15 := bfator[15]:

```
degree( rb15, b);
```

612

(18)

```
#Apply the Möbius transformation on the interval (1, 2 + sqrt(3)), constructing Rb15
```

$$bmoebius := \left\{ b = \frac{1 + (2 + \sqrt{3})d}{1 + d} \right\} :$$

```
Rb15 := collect(numer(expand(subs(bmoebius, rb15))), d) :
```

```
#From here we will analyze the signal variation of all coefficients Rb15
```

```
DescartesSignChanges(Rb15, d);
```

3

(19)

```
#Finding possible candidates for zeros of the factor rb15
```

```
fsolve(rb15=0, b, 1 .. (2 + sqrt(3)));
```

2.018441016, 3.036611183, 3.327892663

(20)

#We now repeat the same process for the resultant of $r1$ with $\tilde{f}$, in relation to b

$R2 := \text{resultant}(r1, \tilde{f}, b) :$

#Compute the factors of the resultant $R2$ and count how many there are

```

dg(a) := proc(f)
  if type(f, '^') then
    degree(op(1,f), a);
  else
    degree(f, a);
  end if;
end proc;

afator := sort(
  [op(factor(R2))],
  (f, g) → dg(a)(f) < dg(a)(g)
) :

```

$nops([op(factor(R2))])$;

15

(21)

#We list below the factors of the resultant $R2$, except for the factor 15 because it is too large. When necessary, we will use Sturm's algorithm to check if the factor cancel each other out within the interval $[1, 2 + \sqrt{3})$ where γ_l is defined

$ra1 := afator[1];$

$ra1 := -206294879089744981779124551495182122153863456743895446897623467852562432$ (22)

$ra2 := afator[2];$

$ra2 := a^{1392}$

(23)

$ra3 := afator[3];$

$ra3 := a^2 + 4a + 1$

(24)

$ra4 := afator[4];$

$\text{sturm}(\text{sturmseq}(ra4, a), a, 2, 5);$ #number of roots in $[2, 5)$

$ra4 := a^2 - 4a + 1$

1

(25)

$ra5 := afator[5];$

$ra5 := (a^2 - 3)^5$

(26)

$ra6 := afator[6];$

$ra6 := (a^2 + 1)^{494}$

(27)

$$\begin{aligned}
ra7 &:= afator[7]; \\
sturm(sturmseq(ra7, a), a, 2, 5); \#number of roots in [2,5) \\
ra7 &:= a^4 - 4a^3 + 18a^2 - 4a + 1 \\
0
\end{aligned} \tag{28}$$

$$\begin{aligned}
ra8 &:= afator[8]; \\
ra8 &:= a^4 + 4a^3 + 18a^2 + 4a + 1
\end{aligned} \tag{29}$$

$$\begin{aligned}
ra9 &:= afator[9]; \\
sturm(sturmseq(ra9, a), a, 2, 5); \#number of roots in [2,5) \\
ra9 &:= (7a^4 - 6a^2 + 3)^5 \\
0
\end{aligned} \tag{30}$$

$$\begin{aligned}
ra10 &:= afator[10]; \\
sturm(sturmseq(ra10, a), a, 2, 5); \#number of roots in [2,5) \\
ra10 &:= 47a^6 - 195a^4 + 141a^2 - 65 \\
0
\end{aligned} \tag{31}$$

$$\begin{aligned}
ra11 &:= afator[11]; \\
sturm(sturmseq(ra11, a), a, 2, 5); \#number of roots in [2,5) \\
ra11 &:= (15a^6 - 3a^4 + 45a^2 - 1)^2 \\
0
\end{aligned} \tag{32}$$

$$\begin{aligned}
ra12 &:= afator[12]; \\
ra12 &:= (17a^6 + 3a^4 + 51a^2 + 1)^2
\end{aligned} \tag{33}$$

$$\begin{aligned}
ra13 &:= afator[13]; \\
sturm(sturmseq(op(1, ra13), a), a, 2, 5); \#number of roots in [2,5) \\
ra13 &:= (175559a^{48} - 1769112a^{46} + 6749652a^{44} + 9460584a^{42} - 193871370a^{40} \\
&\quad + 750687384a^{38} - 111548076a^{36} - 9218817960a^{34} + 37541503329a^{32} \\
&\quad - 86577381616a^{30} + 144267501288a^{28} - 188241754608a^{26} + 198904792788a^{24} \\
&\quad - 173518719696a^{22} + 126058415880a^{20} - 76187388496a^{18} + 37964657145a^{16} \\
&\quad - 15394822392a^{14} + 5027018436a^{12} - 1306370040a^{10} + 287741334a^8 - 59068744a^6 \\
&\quad + 10970148a^4 - 1480584a^2 + 96383)^{10} \\
0
\end{aligned} \tag{34}$$

$$\begin{aligned}
ra14 &:= afator[14]; \\
sturm(sturmseq(op(1, ra14), a), a, 2, 5); \#number of roots in (2,5] \\
ra14 &:= (44405496165211489a^{156} + 778432081743373806a^{154} + 4588983301097861547a^{152} \\
&\quad + 91463528253800665452a^{150} + 1735997934385890809049a^{148} \\
&\quad + 11834475584313712598610a^{146} - 251950730909385181924677a^{144} \\
&\quad - 346879421248819184453976a^{142} + 19687900365414339934332867a^{140} \\
&\quad - 991610283863121627224829598a^{138} - 2330734140754831613723067951a^{136}
\end{aligned}$$

$$\begin{aligned}
& + 43644494250936302528980825740 a^{134} - 1185266340216425882511662940621 a^{132} \\
& + 965498050544660381854646053470 a^{130} + 51827298493210383572604009983769 a^{128} \\
& - 798953306339655826800382819027520 a^{126} \\
& + 2418561486352732770247081063125732 a^{124} \\
& + 22505999118362201449613086293175608 a^{122} \\
& - 291979589000490034255976063805058996 a^{120} \\
& + 1201823407849735701767493862937572080 a^{118} \\
& + 2293620483252666120653229957450482436 a^{116} \\
& - 54968349747308004622619101750438484280 a^{114} \\
& + 298815787919895766023209219153001367116 a^{112} \\
& - 565170501387507477526976272173989191584 a^{110} \\
& - 2853091661016888289000999525164207068516 a^{108} \\
& + 27985760058406148166512064129422041599912 a^{106} \\
& - 123829999220749404322651513548950921755404 a^{104} \\
& + 364131146300339550961244987663105308453808 a^{102} \\
& - 770693666491457720745309482695477487189988 a^{100} \\
& + 1191542222972832359072223854754521997252696 a^{98} \\
& - 1306622591090883817394234393674596223295532 a^{96} \\
& + 880308412623124256747281454141863757631552 a^{94} \\
& + 11346970395126931728531340682395108604478 a^{92} \\
& - 1161007460020240103947662927112728331749916 a^{90} \\
& + 2501060970095357501980378629424664977799466 a^{88} \\
& - 3799336966843999476217846615982967355699608 a^{86} \\
& + 4216464510482355401209665009526725091213582 a^{84} \\
& - 2878653164479224683536112576321112323318628 a^{82} \\
& + 22782768104289547570883952701155905332298 a^{80} \\
& + 2949579172284263875517603053513135217112560 a^{78} \\
& - 4515095823857240315069222923490007949934742 a^{76} \\
& + 4086074928051909692126718458679077736733980 a^{74} \\
& - 2248582796136185023260114303018516675317714 a^{72} \\
& + 93688929788265465678894977082613567396968 a^{70} \\
& + 1439483793322603463170171355171460441997002 a^{68} \\
& - 1889354214789446704832152008454964815010716 a^{66} \\
& + 1397501719614025495696613521508333409952734 a^{64}
\end{aligned}$$

$$\begin{aligned}
& - 501464125075972356709761348946349688967104 a^{62} \\
& - 268751379640310899986208819214716043596780 a^{60} \\
& + 651556703278695497184558623528782089349464 a^{58} \\
& - 671516898313972531371291330569851480619940 a^{56} \\
& + 500660534350972827100611625354864176805808 a^{54} \\
& - 301158198299676427416570645288899746788300 a^{52} \\
& + 152986091428732734077127308639814945162408 a^{50} \\
& - 67223130227783141775368744635172989695524 a^{48} \\
& + 25898918782711300837751466027572834451552 a^{46} \\
& - 8822427556502312342130422113129913421300 a^{44} \\
& + 2671586257758021895076610864274710191560 a^{42} \\
& - 720758027732669146959818597806343051580 a^{40} \\
& + 172995997550938533001785912480586741488 a^{38} \\
& - 36747319342870175142158551198807008500 a^{36} \\
& + 6830149278312126704277975923523008568 a^{34} \\
& - 1083297494058569654128894561687341660 a^{32} \\
& + 138274403647061857259270315687417280 a^{30} \\
& - 11901539844399448942914922276770711 a^{28} \\
& + 10234444613452164094595065416606 a^{26} + 239918939823066018790705974028675 a^{24} \\
& - 54754172262602794250844978956916 a^{22} + 7534724591572525692260253704673 a^{20} \\
& - 652220249958489356161152918622 a^{18} + 18381275079956913389760038355 a^{16} \\
& + 4466827715637094880030762664 a^{14} - 860863706423618339042756309 a^{12} \\
& + 80923719155728268966782482 a^{10} - 4623401410378690841280183 a^8 \\
& + 207351687482767870069356 a^6 - 13154712311724387999525 a^4 \\
& + 773931217904095649454 a^2 - 18927294346712438127)^2
\end{aligned}$$

1

(35)

#Finding possible candidates for zeros of the factor ra14

fsolve(op(1, ra14) = 0, a, 2 .. 5);

3.474957787

(36)

#The factor ra15 has degree 612, so we will not show it, just compute how many zeros it has in the desired range.

ra15 := afator[15];

degree(ra15 , a);

#Apply the Möbius transformation on the interval (2, 5), constructing Ra15

$$amoebius := \left\{ a = \frac{2 + 5 \cdot c}{1 + c} \right\} :$$

Ra15 := collect(numer(expand(subs(amoebius, ra15))), c) :

#From here we will analyze the signal variation of all coefficients Ra15

DescartesSignChanges(Ra15, c);

#Finding possible candidates for zeros of the factor ra15

fsolve(ra15=0, a, 2 .. 5);

3.883374171, 3.888068164

#Let's rule out that the root $a=2+\sqrt{3}$ does not form a pair with any of the other candidates b

#Apply the Möbius transformation on the interval $[1, 2+\sqrt{3})$, constructing $\hat{r}1$

$$moebius := \left\{ b = \frac{1 + (2 + \sqrt{3})d}{1 + d} \right\} :$$

$\hat{r}1 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(moebius, \text{eval}(r1, a = 2 + \sqrt{3}))))), d) :$

#From here we will analyze the signal variation of all coefficients $\hat{r}1$

DescartesSignChanges($\hat{r}1, d$);

(40)

$fsolve(\text{eval}(r1, a = 2 + \sqrt{3}) = 0, b, 1..2 + \sqrt{3})$

(41)